Pierre Fihey

Paris, France | pierre.fihey@ip-paris.fr | 06 58 35 41 08

Education

Institut Polytechnique de Paris, PhD in Computer Science 2025 – ...

- Exploring the geometry of textual representations and the use of statistical tools such as data depth functions for Anomaly Detection.

Institut Polytechnique de Paris, Master 2 Data Sciences 2023 – 2024

- Optimal Transport, Advanced AI for texts and graphs, NLP, Audio and music information retrieval, Optimization for data science...

Paris Dauphine University, Master 2 Statistical and Financial Engineering 2022 – 2023
(Apprenticeship programm)

- *Master thesis*: Financial news studies toautomatize investments using NLP Algorithms.

Paris Dauphine University, Master 1 Applied Mathematics 2021 – 2022

- *Master thesis*: Prediction of corporate fraud using multiple Machine Learning tools.

Paris Dauphine University, Bachelor's degree in Applied Mathematics 2017 – 2021

Experience

Lecturer, Paris Dauphine University - Master 2 Statistical and Financial Engineering 2024-...

- *Course :* Python for Data Science.
- Development of course content in unsupervised and supervised learning, deep learning, and NLP, including a project on automated decision-making in financial markets.

Research Intern in NLP, Telecom Paris - LTCI 2024-2025

- Study of Large Language Models for Anomaly Detection in text data: Unsupervised Learning, Knowledge Distillation, Contrastive Learning, Representation Learning..

Quantitative researcher (Apprenticeship), Ofi Invest Asset Management 2022 - 2023

- Quantitative research projects for investment solutions : Study of financial news using NLP algorithms by combining Name Entity Recognition (NER), Few-Shot Learning and Aspect Based Sentiment Analysis (ABSA) methods.

Publications

[1] Pierre Fihey, Matthieu Labeau, and Pavlo Mozharovskyi. "A Geometry-Aware Depth Function for Robust Unsupervised Textual Anomaly Detection". In: *Findings of the Association for Computational Linguistics: EMNLP 2026 (to be published)*. 2026.

[2] Pierre Fihey, Matthieu Labeau, and Pavlo Mozharovskyi. "Enhancing Two Steps Textual Anomaly Detection through Anisotropy Mitigation". In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2026.

Talks

- **ACL 2026 - Paper Presentation (Oral)** - *Enhancing Two Steps Textual Anomaly Detection through Anisotropy Mitigation*